What can an ethical engineer be but a good person? That's how I'd like to answer. Unfortunately, we don't live in a society where you can survive off your moral compass, which is why this question still needs to be asked. Any philosophical, moral, or ethical topic has a debatable amount of flexibility and the opportunity to twist words and find evidence for anything under the sun. They are inherently complex, subjective, and inconclusive, qualities that have always made me stray away from researching classical thinking. The multi-dimensional nature makes it seem futile to dive deep into these topics, but people never will and never should stop thinking, so it's better to exercise the noggin as much as you can than stay silent or stuck. Whether you do that by conforming your thoughts to classic schools like the big three we have gone over in class, or you try your hardest to freestyle it and make your own conclusions, everyone has a unique way of exploring ethics. I am certain of my own. Which, admittedly, is hypocritical to say in this context, but uncertainty is my number one enemy and the reason I've always pushed against philosophy; I'd like to be sure of *something* when writing an entire think piece.

To better describe the ethical engineer, it's good to understand that it's harder to be a clear-cut UNethical engineer. There aren't many evil scientists out in the wild. You can't single out a company and blame every employee in the organization. There are far too many safeguards in modern professional work for something intentionally malicious to make it into the market. There's a miniscule chance of any obviously bad resolution to a conflict being followed. When considering consequentialism, the ideology I'd think is the most popular in engineering contexts, a lot of the time it's clear just based on instinct what a bad and a good outcome is. Plain consequentialism involves maximizing the "good" of an outcome. We debated a lot about what "good" means in class, but I feel it gets clear as soon as you take a step or two into the situation at hand. Thinking deontologically, it's still clear what good and bad actions are. Killing someone bad, helping someone good, you are obligated to not hurt people, etc. Finally, it's tough to make it as a person, much less in engineering, if the virtues you hold closest are laziness, selfishness, or hostility.

Virtue ethics is an approach to ethical reasoning that evaluates how elements of one's character manifest themselves into ethical behavior. Virtue ethics are an under looked aspect of engineering and my personal favorite. Ethical engineers constantly exemplify the virtues of respect, cooperation, modesty, and integrity in their job without considering it. On the surface, they must value, listen to, and collaborate with their colleagues, do their best work, and speak up when they see tomfoolery. However, as explored by Lloyd and Busby, micro ethical decisions are also involved in endless everyday tasks, and an ethical engineer can implicitly rely on their personal values to make such decisions. It is entirely too tedious to stop and consider every decision you make, so it is comforting to know that your virtues guide your practice.

Virtue ethics also provides necessary leeway for ethical decision making. At the end of the day, engineers must make a choice. Unfortunately, that choice can end up not having great results. In deontological ethics, the theory that morality is based solely on actions and duties, if your actions had good intentions, it was ethical. However, for an empathic individual, this won't sit right with them. How could you be okay with something that harms someone or something else? On the other hand, in plain consequential ethics, the whole situation is boiled down to being unethical and wrong. Being too harsh on yourself leads to burnout, personal issues, and a general disservice to your well-being. Your virtues are something you can rely on through tough times. If you are learning from your mistakes, it doesn't taint the elements of your character, and you can continue to output your positive influence on your surroundings and pursue your passions with a chip on your shoulder.

Consequentialism, the theory that decisions are moral based on their outcomes, is still useful. It is the most applicable to the engineering profession and thus easiest to understand and the easiest way to get an engineer to think about ethics. Consequentialism is attractive due to its reasonable nature; in my opinion the most famous example of consequentialism is common sense. Any sane person can conclude that they shouldn't design or approve a product that has a chance of exploding because that creates negative outcomes. It seems obvious, but an approachable and thorough source like Stanford's article on consequentialism exposes the reader to the complexity of ethical reasoning. It goes over the offshoots of consequentialism that attempt to cover more bases of moral dilemmas and how they stack against each other and compares them to other schools of thought. The article and consequentialism are great introductory resources as well as a showcase of a tried-and-true ethical theory. For me, in this stage, going from a "yeah duh" attitude to a "let me think about this for five seconds" attitude was invaluable.

Factoring yourself into the equation is the hardest part. Your personal relationships, values, and obligations to yourself are all things a human must consider. An ethical engineer can't be narcissistic, but you have to place limits on morals. Stanford's consequentialism article touches on this, citing satisficing consequentialism and progressive consequentialism as two options to reason why you don't always have to do the best thing to be ethical. Agent-centered deontology also focuses on the duties you have to yourself and the special considerations they have in comparison to duties you have to others. At the end of the day, you must provide for yourself. The ideal ethical engineer would find a job they align with, one that exemplifies their values and provides benefit to the world, and contribute to the company's work. Unfortunately, sometimes you are forced into a certain path by your surroundings. Last week I told my grandmother that I'd either be making thermostats or war robots for a living and as much as I kid, that's a reality for a lot of people. Making ninety thousand dollars working for a defense contractor and minorly

contributing to an ultimately unstoppable global phenomena to support oneself, their family and friends, and whoever they can provide utility to by continuing to stay alive must be weighed. Struggling with expectations that you observe or set for yourself ethically calls for a framework to approach growth with.

The study on real life examples of engineering ethics by Fila et al. provides a fantastic example of what I would consider the ethical pipeline. In figure 2, the diagram goes from the bottom left (category A) of accepting and following protocols to the top right (category F) of questioning industry standards, stewarding new culture, and building a better industry ecosystem. Seniority is the main factor in the importance of ethical engineering. As you get older, as you gain wisdom, how can you not start to question what goes on around you? Simultaneously, as you gain seniority in your career, you obtain the ability to question and influence the operation of your daily activities. In the study, the interviewees weren't acutely focused on this concept, but I believe it's the underlying reason as to why the categories exist. Participants that fell into category A determined that following the procedures laid out for them with integrity was the most ethical option. Many were concerned with the outcomes of their work but assumed that appealing to the rules was how the best outcome would manifest. They had people to go to for guidance and people to offload work to. This is what I imagine most entry level positions will force you into even if you have deeper concerns. Category B, ensuring integrity, sounds like the people in charge Category A and can push them towards following procedure well. Climbing to category C, upholding professional responsibility, engineers had the chance to stand against internal pressures and consider their responsibilities to the public. This requires at least some amount of authority that engineers in lower categories don't have. This pattern continues up to category F, stewarding culture, where leaders and innovators consider all aspects of engineering ethics and ensure their organization enables all the engineers within to practice ethical engineering. While experience shouldn't be an excuse to not practice ethical engineering, being aware of potential limitations helps maximize your ethical practice (and provides some reassurance for those who worry).

The first few categories will be very applicable for my peers and I in the coming years. Further understanding of these positions is going to be vital to practicing ethical engineering in the beginning of our careers. Kyle Payne, PH.D., would describe these engineers as ethical followers. In an article summarizing a study interviewing twenty-five engineering professionals and their experiences with ethical dilemmas, he describes several behaviors that can help ethical followers navigate difficult situations. Similarly to the engineers in category A and B in the study by Fila et al., they faced pressure to cut corners, mislead clients, approve work without authority, and provide services outside of their competence. In the Fila et al. study, engineers mentioned offloading work to

engineers with the correct experience; Payne's study describes twelve other behaviors that can guide inexperienced engineers, three of the most relevant being *committing, negotiating,* and *refusing*. *Committing* involves embracing a leader in an informed. Good leaders demonstrate honesty, trust, and accountability. Finding a good role model is essential for an ethical follower. Next, *negotiating* refers to working with a leader to compromise on the leader's and follower's interests. This exemplifies the ability of the follower to question and appeal to a leader. Finally, *refusing* simply means refusing to execute a leader's demands. It is a follower's simplest and most blunt method of standing for their beliefs but carries risk of repercussions (getting fired) [1]. These simple behaviors are engrained in all professional contexts but are particularly relevant in innovative processes like engineering. A deeper understanding of ethical followers' struggles and reactions is simply another route to developing an identity and habits as an ethical engineer.

One last thing to consider is the extent of duties of an engineer. A relevant example to my generation I recently pondered would be developing a Roblox game that essentially farms money off children using in-game transactions. In the gaming industry, a good majority of games rely on microtransactions to be able to make profit, that's how they support the livelihoods of their engineers. As long as the purchases aren't egregious and the game is good, most customers are more than happy with this system. However, Roblox's market is overwhelmingly children, and the advertising of quick progression for few dollars is a component of nearly every game on the platform. The games are simple and repeatable, opening the market to an endless number of small developers, and many games do feel monetarily predatory. There are endless examples of a child playing a game on a device owned by a parent or guardian that has their credit card information saved into it and racking up thousands of dollars of purchases before anyone notices. At the same time, it's not the responsibility of a developer to stop anyone from spending money. There are plenty of consenting individuals with personal finances that don't mind spending a couple of dollars on a game. I'm sure many developers go with the "everyone else is doing it" mindset, which I believe is an applicable mindset to many other situations, but in an ethical context should be a question of "why is everyone else doing it" or "should everyone else be doing it". In this specific situation, it's reasonable to assume the consequences are going to be negligible, but generally an ethical engineer should spend time considering how they can influence their audience's free will in a positive manner.

Outside of any specific quality or thought process, an ethical engineer can take a stance on moral issues. This shows they are educated, concerned, and have the desire to uphold moral standards. The theories we discussed in class are helpful to create a custom process for evaluating moral topics, but not a necessary focus. There are endless

semantics to how you justify your actions and ethical theories are there to help you conclude that what you do is correct or assist you in making tough decisions. In any situation where a real moral dilemma arises, the most educated and caring engineer is the most equipped to handle it. In any situation where the options are cut and dry, the most ethical engineer is the one who thinks about it before acting.

[1] Payne, Kyle. “Engineering Ethics – How to Hold Your Leader Accountable”, https://www.nspe.org/career-growth/pe-magazine/spring-2024/engineering-ethics-how-hold-your-leader-accountable

Side note: company profit was engrained into decision making in many in-class activities. Must ethical engineers spend time considering their position on their current economic system? Argument for another day.